

1. Shared Knapsack

State: Let $U_{\text{sum}} = \sum_{i=1}^n U[i]$.

$DP[i][c][u]$ = maximum Bob-value from items $\{1, \dots, i\}$ with total weight $\leq c$ and total Alice-value exactly u .

Base: $DP[0][c][0] = 0$ for all c ; $DP[0][c][u] = -\infty$ for $u \geq 1$.

Recurrence:

$$DP[i][c][u] = \max \begin{cases} DP[i-1][c][u] & \text{(skip item } i) \\ DP[i-1][c-W[i]][u-U[i]] + V[i] & \text{if } c \geq W[i] \text{ and } u \geq U[i] \end{cases}$$

Answer:

$$\max_{0 \leq u \leq U_{\text{sum}}} \min(u, DP[n][C][u])$$

Implementation: Reverse iteration in c and u eliminates the item dimension: $DP[c][u]$ stores $DP[i-1][c][u]$ for cells not yet updated in iteration i .

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begin
  SHAREDKNAPSACK( $U[1..n]$ ,  $V[1..n]$ ,  $W[1..n]$ ,  $C$ )
   $U_{\text{sum}} \leftarrow \sum_{i=1}^n U[i]$ 
   $DP[0..C][0..U_{\text{sum}}] \leftarrow -\infty$ 
  for  $c \leftarrow 0$  to  $C$  do
     $DP[c][0] \leftarrow 0$ 
  for  $i \leftarrow 1$  to  $n$  do
    for  $c \leftarrow C$  downto  $W[i]$  do
      for  $u \leftarrow U_{\text{sum}}$  downto  $U[i]$  do
         $DP[c][u] \leftarrow \max(DP[c][u], DP[c-W[i]][u-U[i]] + V[i])$ 
    ans  $\leftarrow 0$ 
    for  $u \leftarrow 0$  to  $U_{\text{sum}}$  do
      if  $DP[C][u] \neq -\infty$  then
        ans  $\leftarrow \max(\text{ans}, \min(u, DP[C][u]))$ 
    return ans

```

Runtime:

DP fill: $n \times C \times U_{\text{sum}}$ states, $O(1)$ each

Answer extraction: $O(U_{\text{sum}})$

$$O(n \cdot C \cdot U_{\text{sum}}), \quad U_{\text{sum}} = \sum_{i=1}^n U[i]$$

2. Longest Rapidly Increasing Subsequence

State: $DP[i]$ = length of the longest rapidly increasing subsequence ending at index i .

Base: $DP[i] = 1$ for all i .

Recurrence:

$$DP[i] = 1 + \max\{ DP[j] : 1 \leq j < i \text{ and } 2 \cdot A[j] + 3 < A[i] \}$$

(If no such j exists, $DP[i] = 1$.)

Answer: $\max_{1 \leq i \leq n} DP[i]$.

```

begin
  LRIS( $A[1..n]$ )
  for  $i \leftarrow 1$  to  $n$  do
     $DP[i] \leftarrow 1$ 
    for  $j \leftarrow 1$  to  $i - 1$  do
      if  $2 \cdot A[j] + 3 < A[i]$  and  $DP[j] + 1 > DP[i]$  then
         $DP[i] \leftarrow DP[j] + 1$ 
    return  $\max(DP[1], \dots, DP[n])$ 

```

Runtime:

$$\sum_{i=1}^n (i-1) = \frac{n(n-1)}{2}$$

$$O(n^2)$$

3. Dynamic Programming Tables

(a)

$$A[i, j] = \begin{cases} i + j & \text{if } i = 1 \text{ or } j = n \\ A[i-1, j] + A[i, j+1] & \text{otherwise} \end{cases}$$

Base cases ($n = 4$, shaded = base):

	$j=1$	$j=2$	$j=3$	$j=4$
$i=1$	2	3	4	5
$i=2$	.	.	.	6
$i=3$	.	.	.	7
$i=4$	.	.	.	8

$$A[i, j] \leftarrow \underbrace{A[i-1, j]}_{\uparrow} + \underbrace{A[i, j+1]}_{\leftarrow}$$

Fill $i = 1 \rightarrow n$ ($\downarrow$), $j = n \rightarrow 1$ ($\leftarrow$):

```

begin
  for  $i \leftarrow 1$  to  $n$  do
    for  $j \leftarrow n$  downto 1 do
      if  $i = 1$  or  $j = n$  then
         $A[i, j] \leftarrow i + j$ 
      else
         $A[i, j] \leftarrow A[i-1, j] + A[i, j+1]$ 

```

When computing $A[i, j]$ for $i \geq 2, j \leq n-1$:

- $A[i-1, j]$: already computed in outer iteration $i-1$.
- $A[i, j+1]$: already computed earlier in the current iteration since the inner loop goes $n, n-1, \dots$

Completed table ($n = 4$):

	$j=1$	$j=2$	$j=3$	$j=4$
$i=1$	2	3	4	5
$i=2$	15	13	10	6
$i=3$	45	30	17	7
$i=4$	100	55	25	8

Spot check: $A[4, 1] = A[3, 1] + A[4, 2] = 45 + 55 = 100$.

(b)

$$A[i, j] = \begin{cases} i + j & \text{if } i = 1, i = 2, j = 1, \text{ or } j = n \\ A[i-2, j+1] + A[i+1, j-1] & \text{otherwise} \end{cases}$$

($n = 4$, shaded = base, $\times$ = references outside array):

	$j=1$	$j=2$	$j=3$	$j=4$
$i=1$	2	3	4	5
$i=2$	3	4	5	6
$i=3$	4	9	$\times$	7
$i=4$	5	$\times$	$\times$	8

$A[3, 2] = A[1, 3] + A[4, 1] = 4 + 5 = 9$. $A[3, 3] = A[1, 4] + A[4, 2]$: requires $A[4, 2]$, which references row 5.

Impossible: $\forall n \geq 3$: cell $(n, 2)$ is not a base case ($n \geq 3, j = 2 \neq 1, j = 2 < n$). The recurrence gives:

$$A[n, 2] = A[n-2, 3] + \underbrace{A[n+1, 1]}_{\notin A[1..n, 1..n]}$$

$A[n+1, 1]$ lies outside $A[1..n, 1..n]$. The formula cannot be evaluated.

For $n = 4$: $A[4, 2] = A[2, 3] + A[5, 1]$ and $A[4, 3] = A[2, 4] + A[5, 2]$, both referencing row 5. $A[3, 3] = A[1, 4] + A[4, 2]$ is also impossible since it depends on $A[4, 2]$. $\square$

(c)

$$A[i, j] = \begin{cases} i + j & \text{if } i = 1, i = n, \text{ or } j = 1 \\ A[i-1, j] + A[i-1, j-1] + A[i+1, j] & \text{otherwise} \end{cases}$$

($n = 4$, shaded = base, $\odot$ = cyclic):

	$j=1$	$j=2$	$j=3$	$j=4$
$i=1$	2	3	4	5
$i=2$	3	$\odot$	$\odot$	$\odot$
$i=3$	4	$\odot$	$\odot$	$\odot$
$i=4$	5	6	7	8

Impossible: $\forall n \geq 4$: cells $(2, 2)$ and $(3, 2)$ are both non-base.

Column $j = 2$:

$$\begin{aligned} A[2, 2] &= \underbrace{A[1, 2]}_{\text{base}} + \underbrace{A[1, 1]}_{\text{base}} + A[3, 2] = 5 + A[3, 2] \\ A[3, 2] &= A[2, 2] + \underbrace{A[2, 1]}_{\text{base}} + \underbrace{A[4, 2]}_{\text{base}} = A[2, 2] + 3 + 6 = A[2, 2] + 9 \end{aligned}$$

Substituting: $A[2, 2] = 5 + (A[2, 2] + 9) = A[2, 2] + 14 \implies 0 = 14$. Contradiction.

$\forall j \geq 2$: every column has the same pattern, $A[2, j] \leftrightarrow A[3, j]$ form a cycle. $\square$